

SQL Cheat Sheet

Query order, joins, aggregation, window functions and practical query patterns.

Query Structure

Topic	Quick reference
Select	SELECT col1, col2 FROM table
Filter	WHERE status = 'active' AND amount > 100
Sort/limit	ORDER BY created_at DESC LIMIT 100
Execution order	FROM -> WHERE -> GROUP BY -> HAVING -> SELECT -> ORDER BY

Joins

Topic	Quick reference
INNER JOIN	Rows matching in both tables
LEFT JOIN	All left rows + matching right rows
Anti join	LEFT JOIN b ON ... WHERE b.id IS NULL
Union	UNION removes duplicates; UNION ALL keeps all rows

Aggregation

Topic	Quick reference
Group by	SELECT city, COUNT(*) FROM users GROUP BY city
Having	HAVING COUNT(*) > 10
Common metrics	COUNT(*), COUNT(DISTINCT user_id), SUM, AVG, MIN, MAX

Window Functions

Topic	Quick reference
Row number	ROW_NUMBER() OVER(PARTITION BY user_id ORDER BY created_at DESC)
Lag/lead	LAG(amount) OVER(PARTITION BY user_id ORDER BY month)
Running total	SUM(amount) OVER(PARTITION BY user_id ORDER BY date)

Date & Text

Topic	Quick reference
Dates	DATE_TRUNC('month', created_at), EXTRACT(year FROM date)
Case	CASE WHEN amount > 1000 THEN 'High' ELSE 'Low' END
Nulls	COALESCE(col, 0), NULLIF(a,b)